

VLSI Devices

Lecture 18

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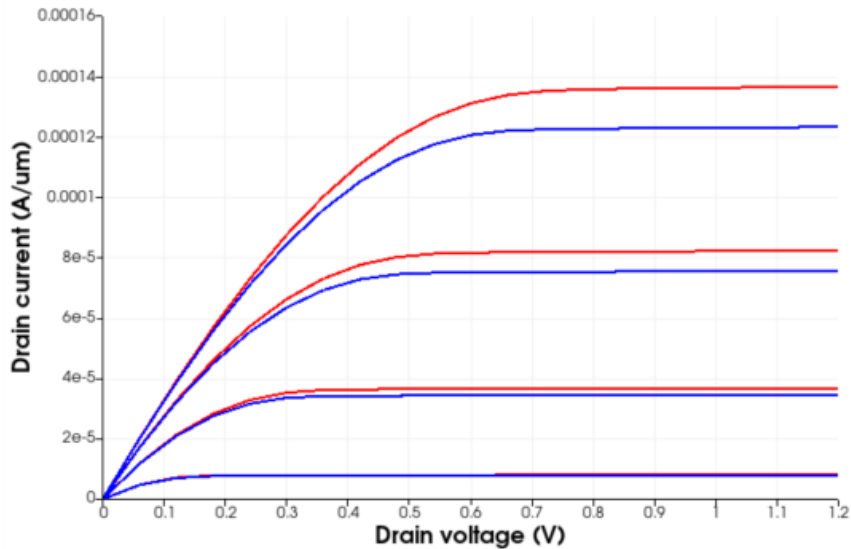
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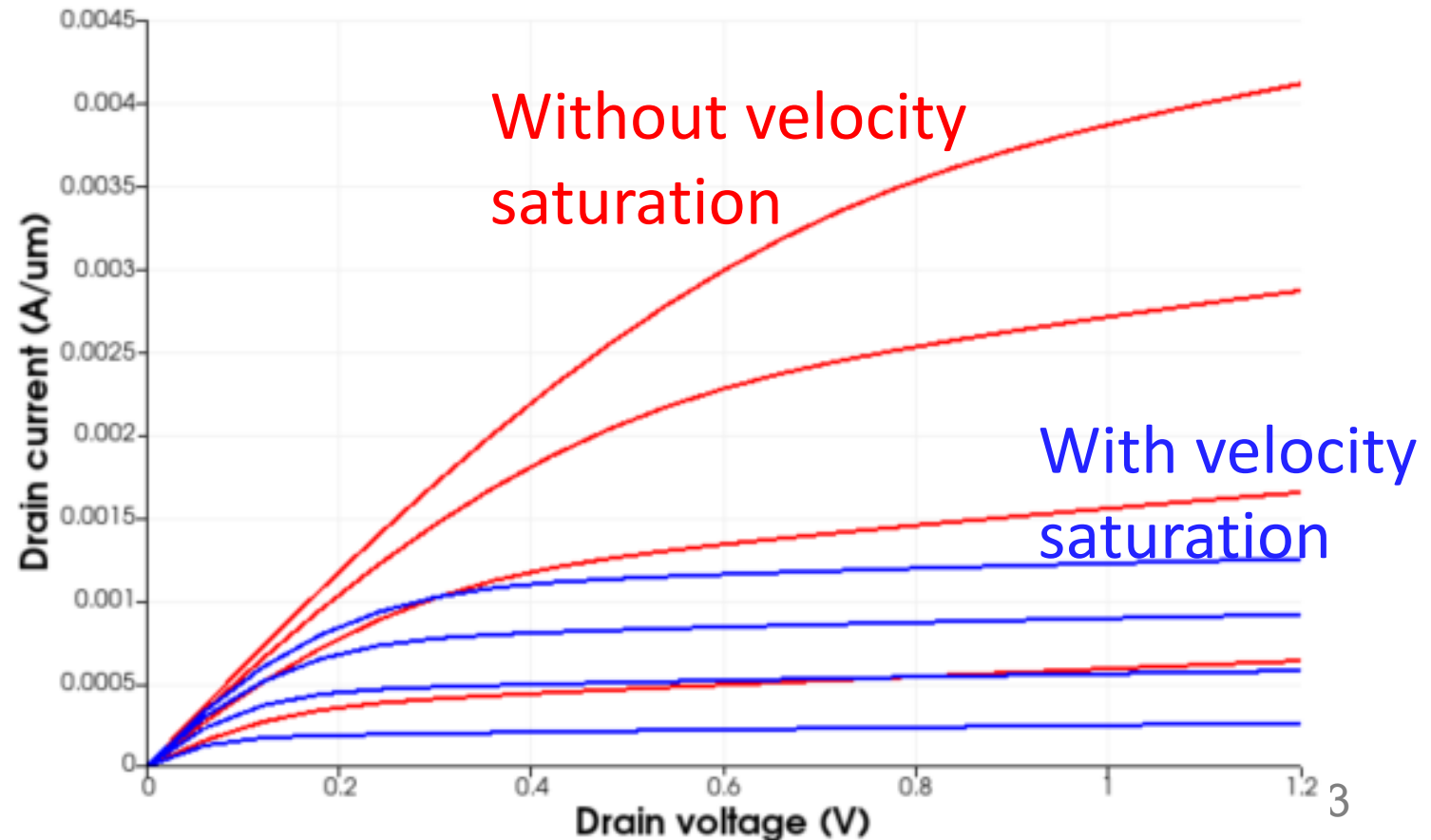
L18

Velocity saturation

- Impact of velocity saturation
 - Saturation occurs at a much lower voltage (than $V_{dsat} = (V_{gs} - V_t)/m$).



Long-channel



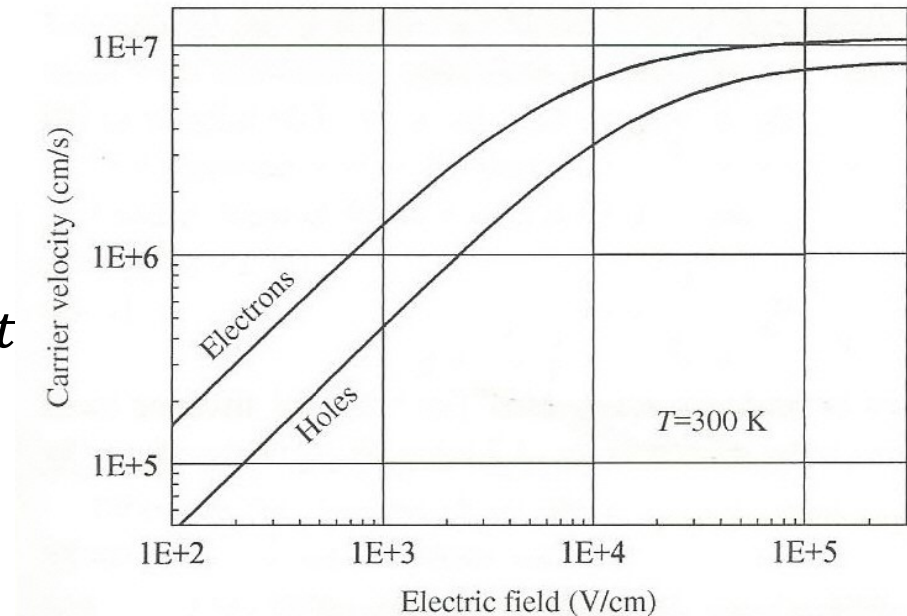
Velocity-field relationship

- Caughey-Thomas

$$v = \frac{\mu_{eff}\mathcal{E}}{[1 + (\mathcal{E}/\mathcal{E}_c)^n]^{1/n}}$$

Taur, Eq. (6.39)

- Critical field, $\mathcal{E}_c$
- For electrons, $n = 2$. For holes, $n = 1$
- At low fields, $v = \mu_{eff}\mathcal{E}$
- At high fields ($\mathcal{E} \rightarrow \infty$), $v \rightarrow \mu_{eff}\mathcal{E}_c = v_{sat}$



Velocity-field relationship
(Taur, Fig. 2.11)

Analytic solution for $n = 1$ (1)

- Valid for holes (PMOSFET)

$$I_d = -W Q_i(V) \frac{\mu_{eff} \frac{dV}{dy}}{1 + \left(\frac{\mu_{eff}}{v_{sat}} \right) \frac{dV}{dy}}$$

Taur, Eq. (6.42)

–Rearranging

$$I_d \left[1 + \left(\frac{\mu_{eff}}{v_{sat}} \right) \frac{dV}{dy} \right] = -W Q_i(V) \mu_{eff} \frac{dV}{dy}$$

$$\textcolor{red}{I_d} = - \left[\mu_{eff} W Q_i(V) + \left(\frac{\mu_{eff} \textcolor{red}{I_d}}{v_{sat}} \right) \right] \frac{dV}{dy}$$

Taur, Eq. (6.43)

Analytic solution for $n = 1$ (2)

- Drain current with velocity saturation

$$I_d dy = - \left[\mu_{eff} W Q_i(V) + \left(\frac{\mu_{eff} I_d}{v_{sat}} \right) \right] dV$$

– Integration from $y = 0$ to L (from $V = 0$ to V_{ds})

$$I_d L = -\mu_{eff} W \int_0^{V_{ds}} Q_i(V) dV - \left(\frac{\mu_{eff} I_d}{v_{sat}} \right) V_{ds}$$

$$I_d L \left(1 + \frac{\mu_{eff} V_{ds}}{v_{sat} L} \right) = -\mu_{eff} W \int_0^{V_{ds}} Q_i(V) dV$$

$$I_d = \frac{-\mu_{eff} (W/L) \int_0^{V_{ds}} Q_i(V) dV}{1 + (\mu_{eff} V_{ds} / v_{sat} L)}$$

Taur, Eq. (6.44)

Analytic solution for $n = 1$ (3)

- Using the charge-sheet model

$$Q_i = -C_{ox}(V_{gs} - V_t - mV) \quad \text{Taur, Eq. (6.45)}$$

$$I_d = \frac{\mu_{eff} C_{ox} (W/L) \left[(V_{gs} - V_t) V_{ds} - \frac{m}{2} V_{ds}^2 \right]}{1 + (\mu_{eff} V_{ds} / v_{sat} L)} \quad \text{Taur, Eq. (6.46)}$$

– By solving $\frac{dI_d}{dV_{ds}} = 0$ at V_{dsat} ,

$$0 = \frac{(V_{gs} - V_t) - mV_{dsat}}{1 + (\mu_{eff} V_{dsat} / v_{sat} L)} - \frac{(V_{gs} - V_t) V_{dsat} - \frac{m}{2} V_{dsat}^2}{\left[1 + (\mu_{eff} V_{dsat} / v_{sat} L) \right]^2} (\mu_{eff} / v_{sat} L)$$

Analytic solution for $n = 1$ (4)

- Manipulation

$$\begin{aligned} & [(V_{gs} - V_t) - mV_{dsat}] [1 + (\mu_{eff} V_{dsat} / v_{sat} L)] \\ &= \left[(V_{gs} - V_t) V_{dsat} - \frac{m}{2} V_{dsat}^2 \right] (\mu_{eff} / v_{sat} L) \\ & \quad 2(V_{gs} - V_t) / m \end{aligned}$$

$$V_{dsat} = \frac{2(V_{gs} - V_t) / m}{1 + \sqrt{1 + 2\mu_{eff}(V_{gs} - V_t) / (mv_{sat}L)}} \leq (V_{gs} - V_t) / m$$

Taur, Eq. (6.48)

$$-L \rightarrow \infty, V_{dsat} = (V_{gs} - V_t) / m$$

$$-L \rightarrow 0,$$

$$V_{dsat} = \sqrt{\frac{2(V_{gs} - V_t)v_{sat}L}{\mu_{eff}m}}$$

Analytic solution for $n = 1$ (5)

- Two extreme cases

- $L \rightarrow \infty$,

$$I_{dsat} = \mu_{eff} C_{ox} \frac{W}{L} \frac{(V_{gs} - V_t)^2}{2m} \quad \text{Taur, Eq. (6.50)}$$

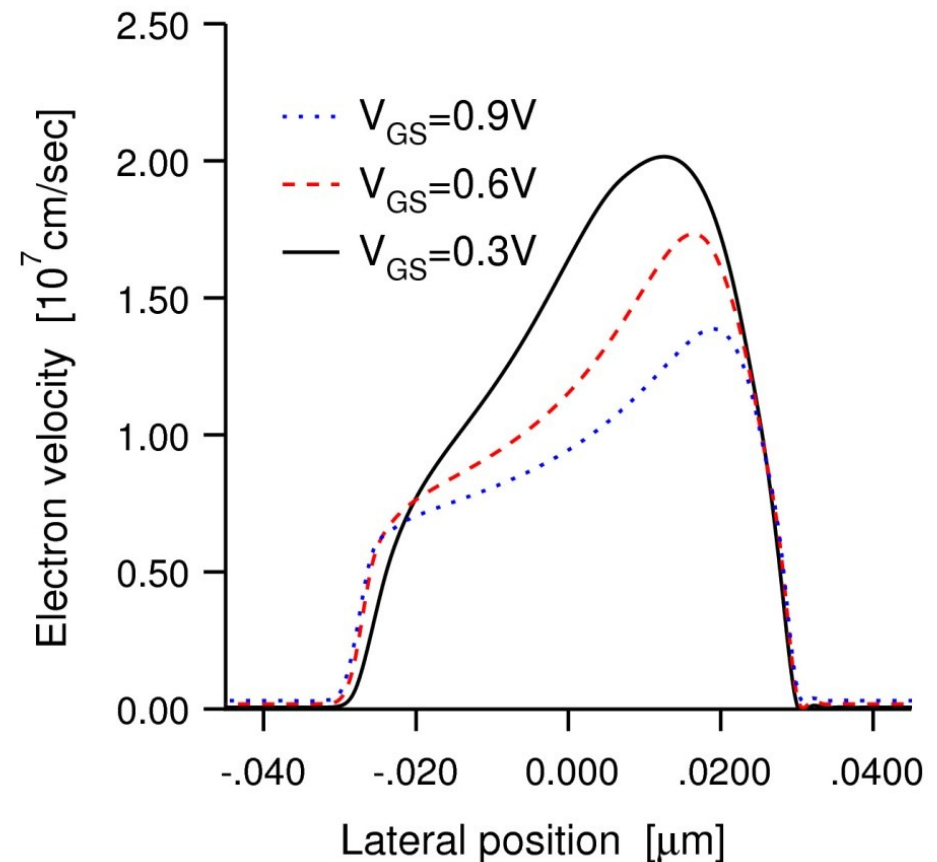
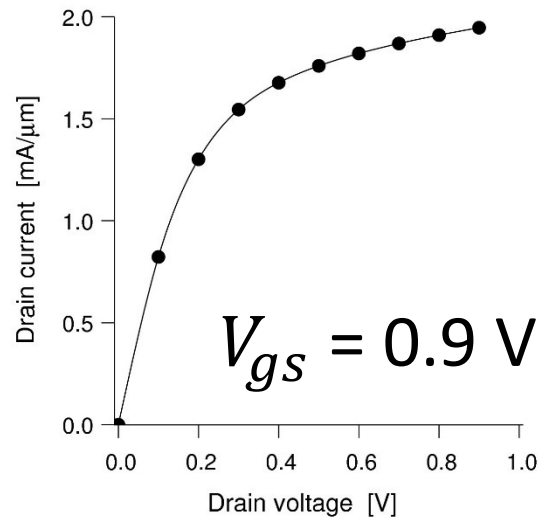
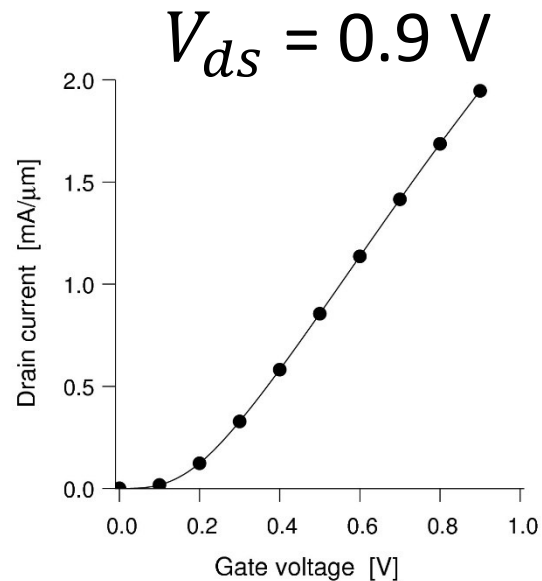
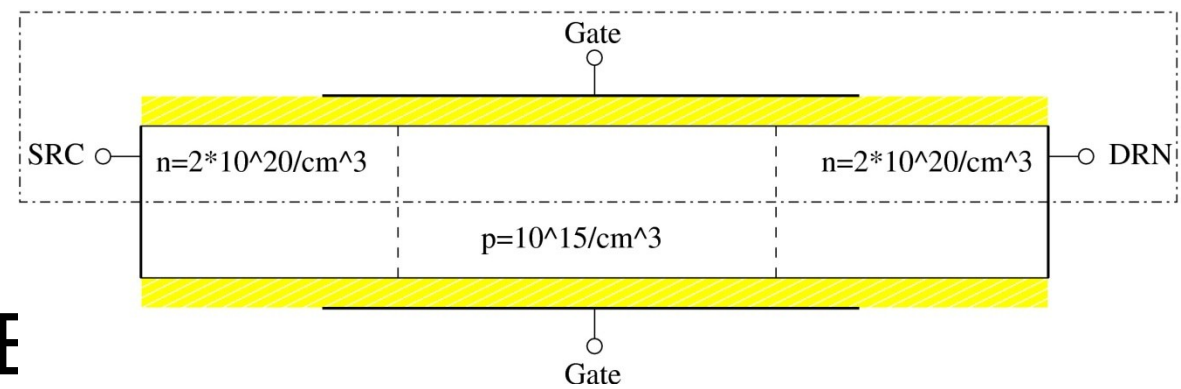
- $L \rightarrow 0$,

$$I_{dsat} = C_{ox} W v_{sat} (V_{gs} - V_t) \quad \text{Taur, Eq. (6.51)}$$

- In this case, I_{dsat} is independent of channel length L and varies linearly with $V_{gs} - V_t$ instead of quadratically as in the long-channel case.
 - Although details are different, different models share two extremes.

Velocity overshoot

- 70-nm-long double-gate MOSFE
 - Strong velocity overshoot



Case study: TSMC 3 nm node

- IEDM 2022 (27.1)

– DIBL versus L_g

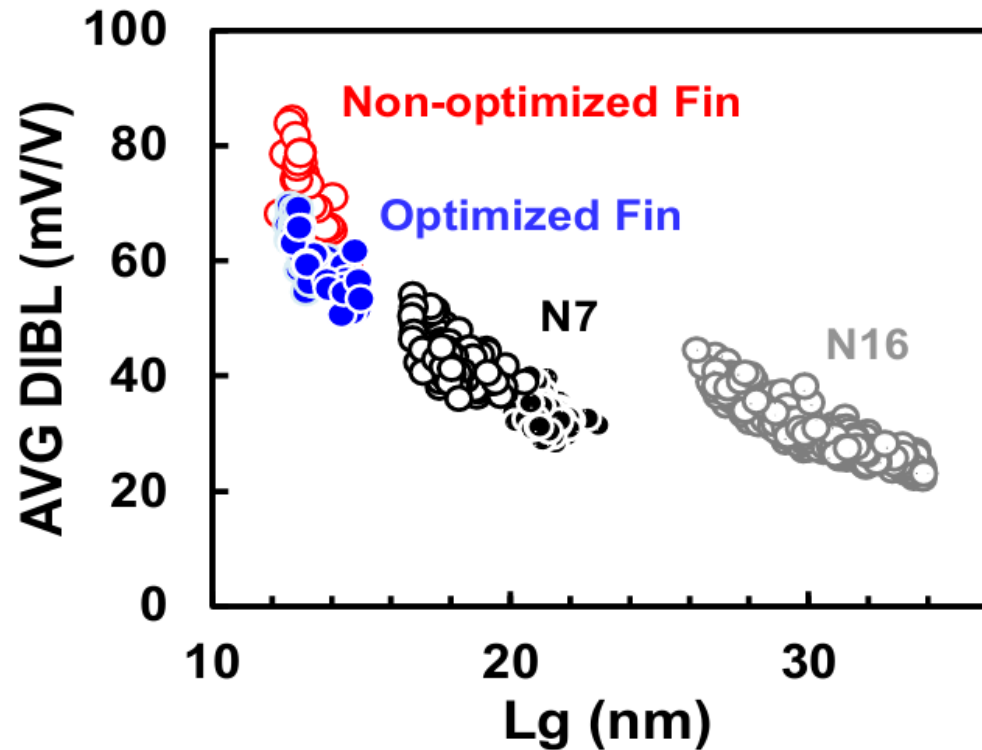


Fig.1 FinFET L_g scaling trend vs DIBL. Fin profile optimization is critical but is at the limit for further L_g scaling.

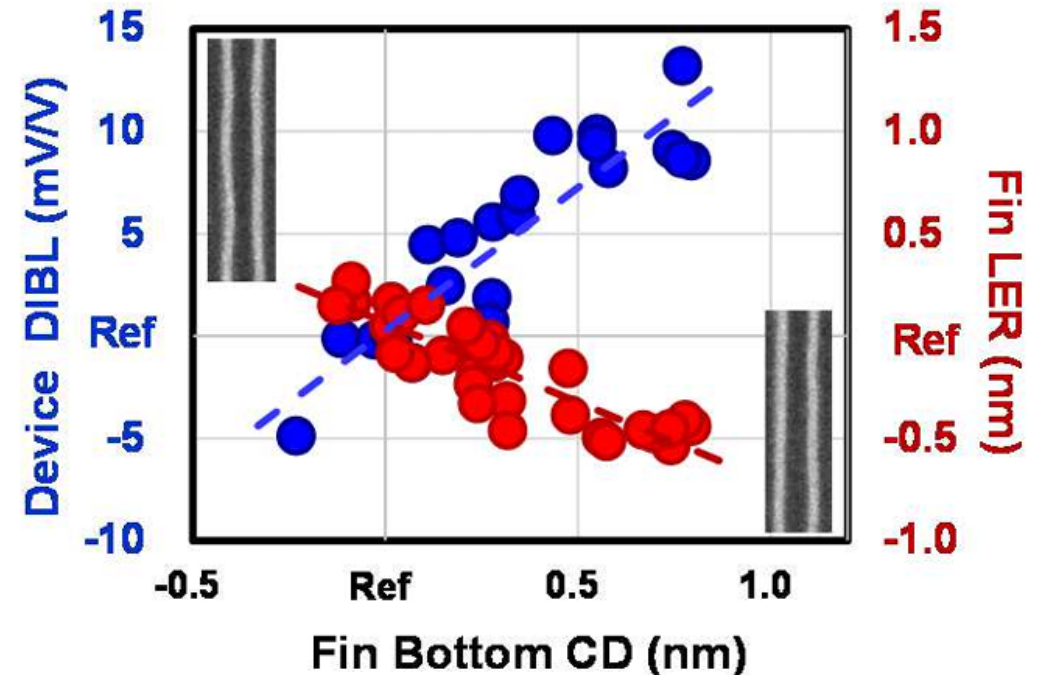


Fig.2 Smaller fin bottom CD reduces DIBL but degrades LER, which is an indicator of fin structural robustness and potential yield impact.

Source-drain series resistance

- Finite silicon resistivity + metal contact resistance
 - MOSFET channel resistance in the linear region

$$R_{ch} = \frac{V_{ds}}{I_d} = \frac{L}{\mu_{eff} C_{inv} W (V_{gs} - V_{on})}$$

Taur, ~Eq. (8.15)

Scaled contacted gate pitch

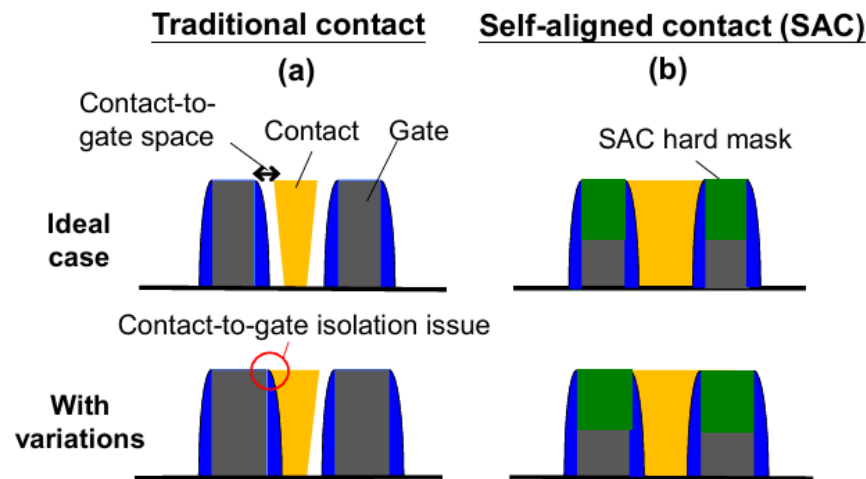
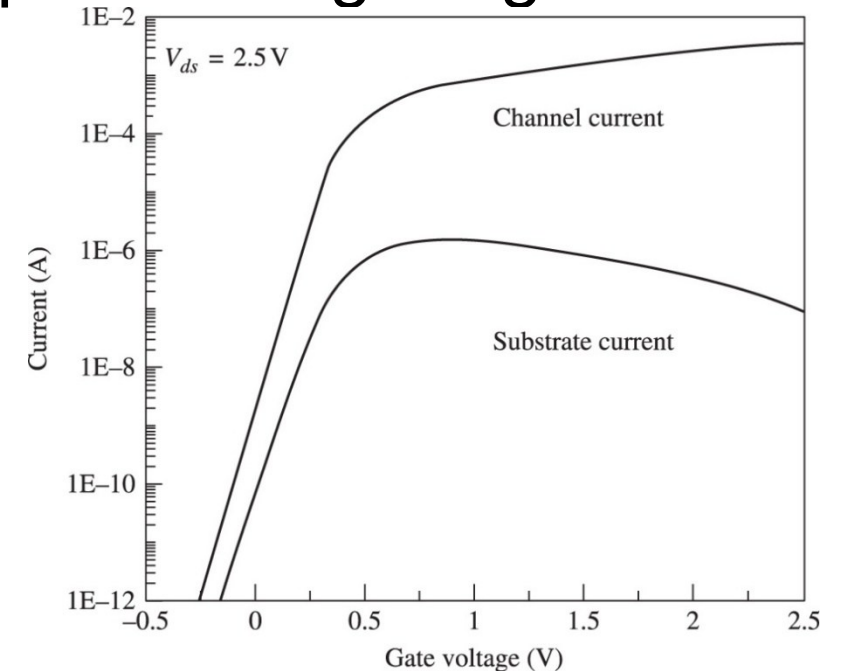
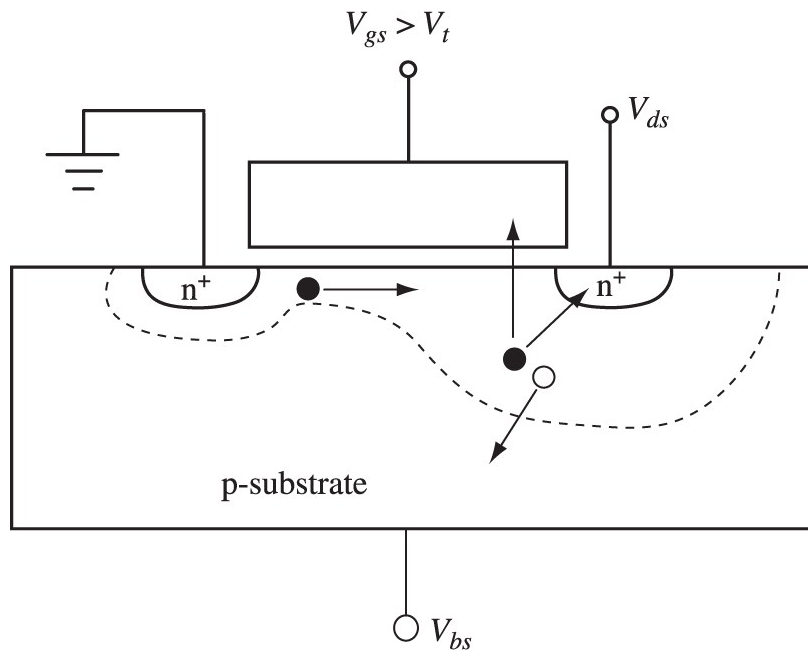


Fig.3 Contact schematics. Traditional contact (a) is vulnerable to variations induced contact-to-gate isolation issues compared to SAC (b).

Self-aligned contact (TSMC, IEDM 2022)

Hot carrier effects

- Electrons gain energy from the electric field.
 - Injection into the SiO_2 layer. Secondary electron/hole pair.
 - $V_{ds} - V_{dsat}$ is the voltage drop in the space-charge region.

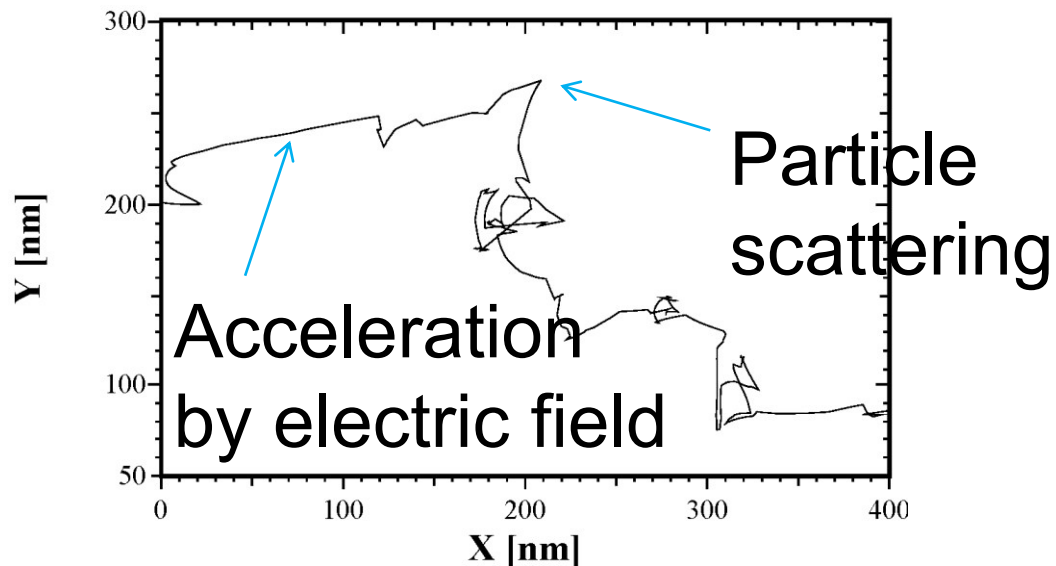


Channel hot-electron (Taur, Fig. 6.30)

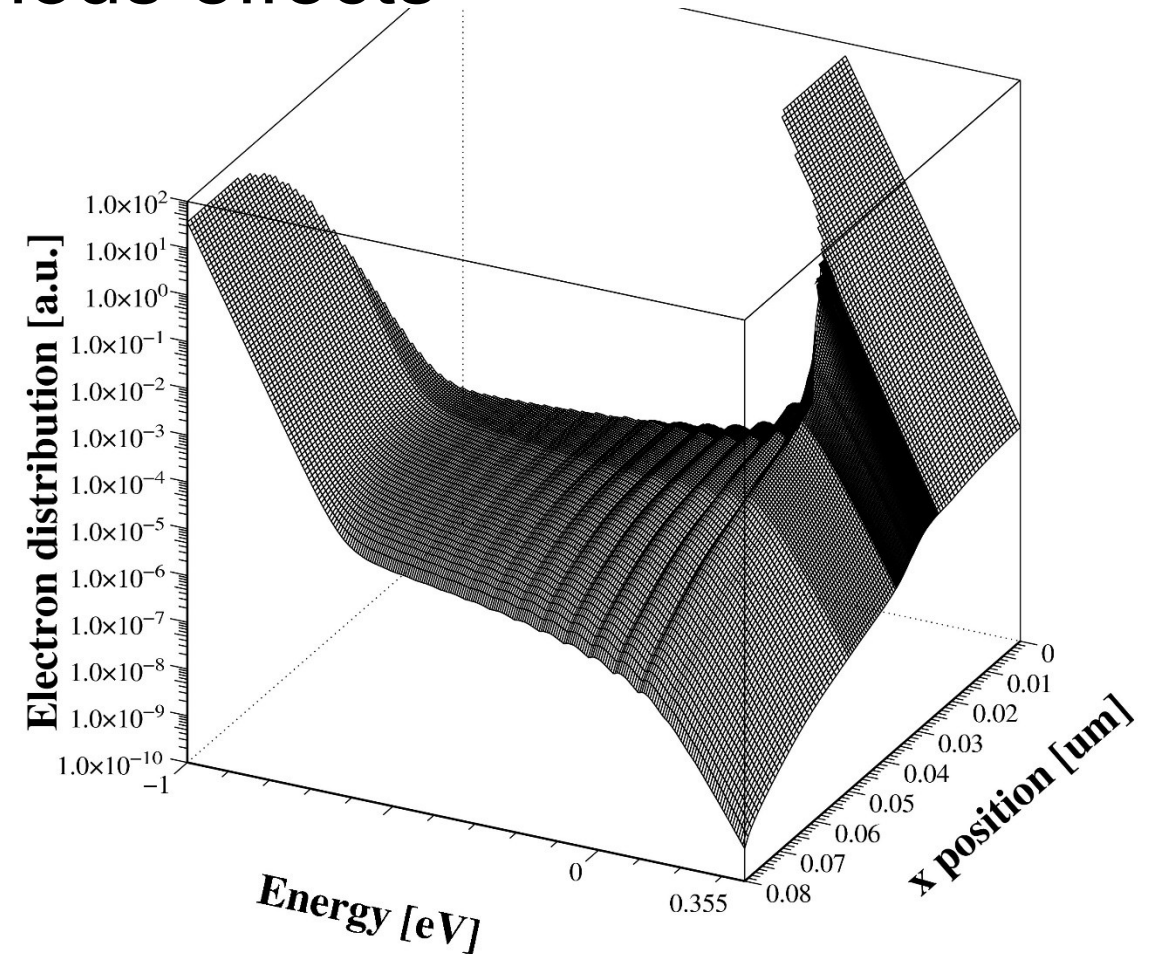
Substrate current (Taur, Fig. 6.31)

Energy distribution, $f(\mathbf{r}, E)$

- Key quantity to understand various effects
 - Various ways to model it
 - Boltzmann transport equation

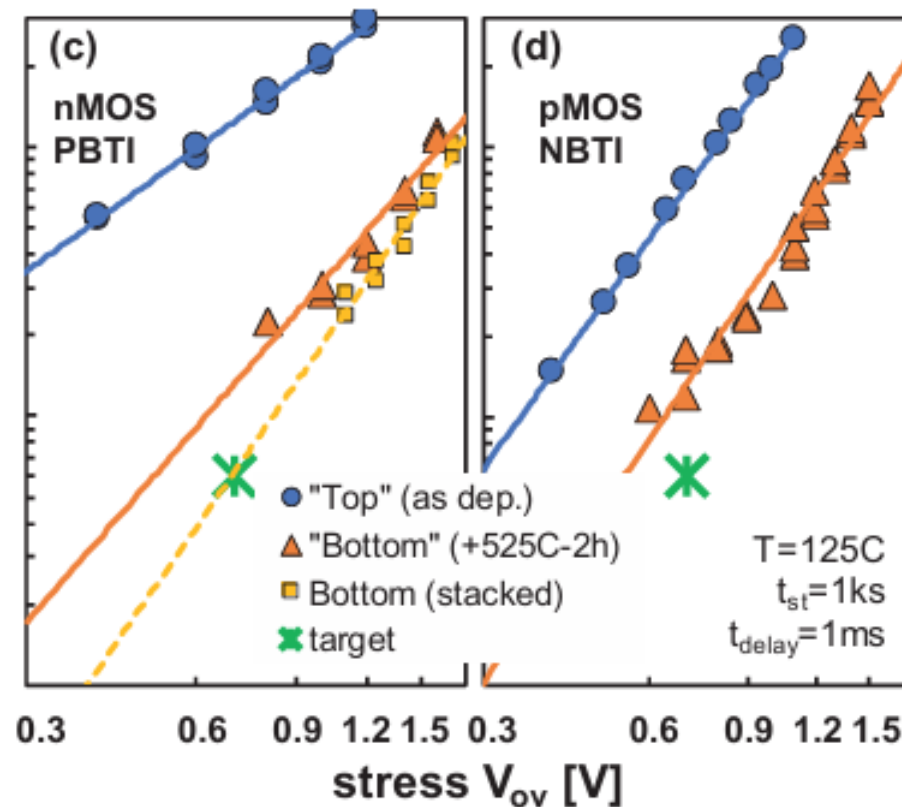


Stochastic electron motion
simulated by Monte Carlo



BTI (Bias-Temperature Instability)

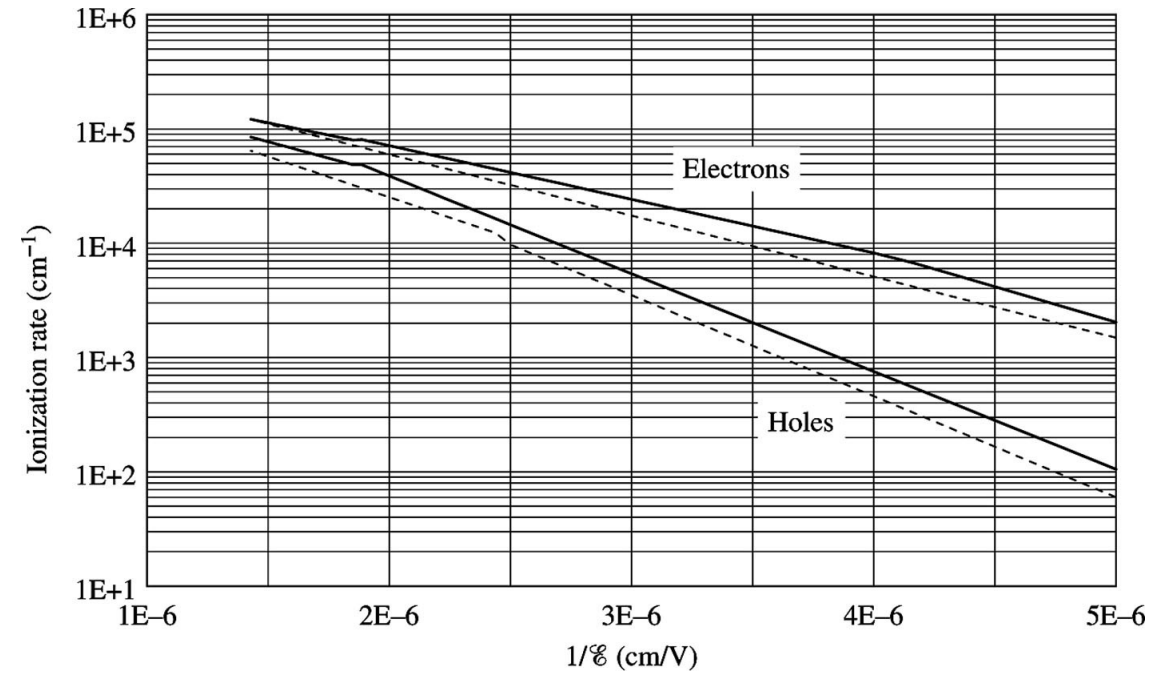
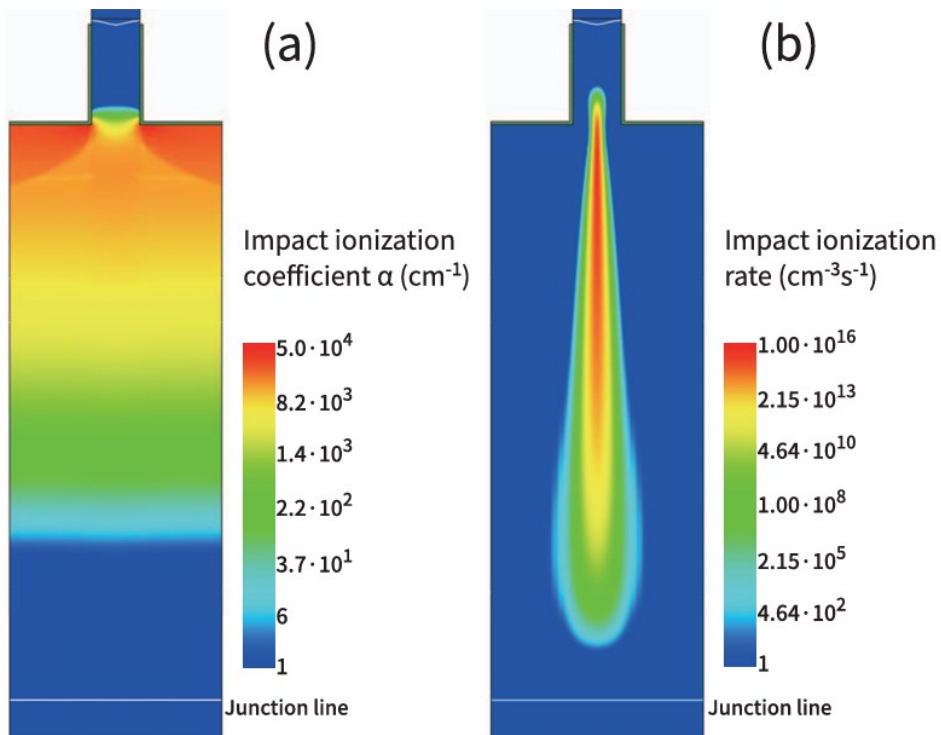
- Reliability issue (NBTI in PMOS, PBTI in NMOS)
 - Interface trap generation & $|V_t|$ shift



BTI characteristics of gate stacks (IMEC, IEDM 2018)

MOSFET breakdown

- Impact ionization
 - Strong dependence on ε
- $$\alpha = A \exp\left(-\frac{b}{\varepsilon}\right) \text{ Taur, Eq. (3.140)}$$



Impact ionization rates
in silicon (Taur, Fig. 3.23)

Thank you!